1.2 Julia Programming Basics

Column vector a is defined as

```
a=[1.0, 2.0 ,3.0]
```

$$\vec{a} = \overset{3 \times 1}{\begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}}$$

The comma in between the numbers tells Julia and this is a column vector.

If you type in

```
a=[1 2 3 4]
```

$$[a] = \overset{1 \times 4}{[1 \quad 2 \quad 3 \quad 4]}$$

Julia will see *a* as a 1 x 4 (1 row by 4 column) matrix, i.e. a row matrix (of integers in this case).

```
b=[2.0 3.0 4.0;8.0 9.0 4.0]
```

The semi colon separates the rows, [b] is a 2 x 3 matrix of float64

$$[b] = \begin{bmatrix} 2.0 & \boxed{3.0} & 4.0 \\ 8.0 & \boxed{9.0} & 4.0 \end{bmatrix}$$

```
b[1,:]
```

$$b[1, :] = \begin{Bmatrix} 2.0 \\ 3.0 \\ 4.0 \end{Bmatrix}$$

```
b[:,2]
```

$$b[: ,2] = \begin{Bmatrix} 3.0 \\ 9.0 \end{Bmatrix}$$

```
xrange=-1:1:1
yrange=1:1:3
loss_values=[x+y^2 for x in xrange, y in yrange]
```

$$\vec{x} = \begin{Bmatrix} -1 \\ 0 \\ 1 \end{Bmatrix}$$

$$\vec{y} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

$$\begin{bmatrix} x_1 + y_1^2 & x_1 + y_2^2 & x_1 + y_3^2 \\ x_2 + y_1^2 & x_2 + y_2^2 & x_2 + y_3^2 \\ x_3 + y_1^2 & x_3 + y_2^2 & x_3 + y_3^2 \end{bmatrix} = \begin{bmatrix} 0 & 3 & 8 \\ 1 & 4 & 9 \\ 2 & 5 & 10 \end{bmatrix}$$

```
a=[1,2,3]
b=[2.0 3.0 4.0;8.0 9.0 4.0]
b*a
```

$$\vec{a} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

$$[b] = \begin{bmatrix} 2.0 & 3.0 & 4.0 \\ 8.0 & 9.0 & 4.0 \end{bmatrix}$$

$$b * a = \underset{2\times 3}{\begin{bmatrix} 2.0 & 3.0 & 4.0 \\ 8.0 & 9.0 & 4.0 \end{bmatrix}} \underset{3\times 1}{\begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}} = \underset{2\times 1}{\begin{Bmatrix} 20 \\ 38 \end{Bmatrix}}$$

$\boxed{\texttt{a.*a}}$

$$\vec{a} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

$$a.*a = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix} .* \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 4 \\ 9 \end{Bmatrix}$$

$\boxed{\texttt{b.*b}}$

$$[b] = \begin{bmatrix} 2.0 & 3.0 & 4.0 \\ 8.0 & 9.0 & 4.0 \end{bmatrix}$$

$$b.*b = \begin{bmatrix} 2.0 & 3.0 & 4.0 \\ 8.0 & 9.0 & 4.0 \end{bmatrix} .* \begin{bmatrix} 2.0 & 3.0 & 4.0 \\ 8.0 & 9.0 & 4.0 \end{bmatrix}$$

$$= \begin{bmatrix} 4.0 & 9.0 & 16.0 \\ 64.0 & 81.0 & 16.0 \end{bmatrix}$$

```
a=[1.0, 2.0 ,3.0]
f(x)=x^3
f.(a)
```

$$\vec{a} = \begin{Bmatrix} 1 \\ 2 \\ 3 \end{Bmatrix}$$

$$f.(a) = \begin{Bmatrix} 1^3 \\ 2^3 \\ 3^3 \end{Bmatrix} = \begin{Bmatrix} 1 \\ 8 \\ 27 \end{Bmatrix}$$

Means operate on every element of a